

Kumar Akash

Ph.D. Candidate, School of Mechanical Engineering, Purdue University

Summary

Innovative control engineer with varied experience in multi-disciplinary research projects. Experienced in modeling, estimation, and control with a focus on human-machine interactions. Specialization in human subject studies and machine learning algorithms to infer human cognitive states using behavioral and psycho-physiological data.

Contact Information

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Education

Ph.D.	Mechanical Engineering, Purdue University, West Lafayette, IN <i>Specialization:</i> Systems, Measurements & Controls <i>Advisor:</i> Dr. Neera Jain <i>Thesis:</i> Reimagining Human-Machine Interactions through Trust-based Feedback	Expected Aug 2020 GPA: 4.00/4.00
M.S.	Mechanical Engineering, Purdue University, West Lafayette, IN <i>Advisor:</i> Dr. Neera Jain	May 2018 GPA: 3.93/4.00
B.Tech.	Mechanical Engineering, Indian Institute of Technology, Delhi, India <i>Advisor:</i> Dr. Sudipto Mukherjee <i>Thesis:</i> Growth Plate Preserving Intramedullary Nail for Pediatric Patients	May 2015 GPA: 9.09/10.00

Research Interests

Deterministic and Probabilistic Dynamic Modeling; Optimal Control; Human-Machine Interactions; Machine Learning

Relevant Coursework

Human Factors in Engineering, Machine Learning, System Identification, Multidisciplinary Design Optimization, Optimal Control & Estimation, Adaptive Control, Nonlinear Feedback Control, Hybrid Systems

Awards and Recognition

- **Bilsland Dissertation Fellowship**, Purdue University, Aug 2019. Awarded by the Dean of the Graduate School to provide support to outstanding Ph.D. candidates in the final year of doctoral degree completion.
- **Batch of Sixty Seven (BOSS) Award**, IIT Delhi, May 2015. Awarded for the best experimental project in mechanical engineering discipline submitted during the session 2014–2015.
- **Summer Undergraduate Research Award**, IIT Delhi, May 2013. Awarded by Industrial Research and Development Unit of IIT Delhi for exceptional research potential displayed at the undergraduate level.

Skills

- Software: MATLAB, Simulink, NI LabVIEW (Real-Time & FPGA), SolidWorks, Unreal Engine
- Languages: Python, C++, JavaScript, LaTeX

Publications

Peer Reviewed Journal Articles

- J1 **K. Akash**, G. McMahon, T. Reid, and N. Jain, “Human Trust-based Feedback Control: Dynamically varying automation transparency to optimize human-machine interactions.” *IEEE Control Systems Magazine*. (Under Review)
- J2 **K. Akash**, W.-L. Hu, N. Jain, and T. Reid, “A Classification Model for Sensing Human Trust in Machines Using EEG and GSR.” *ACM Transactions on Interactive Intelligent Systems*, vol. 8, no. 4, article 27, November 2018. doi: 10.1145/3132743
- J3 W.-L. Hu, **K. Akash**, T. Reid, and N. Jain, “Computational Modeling of Human Trust in Human-Machine Interactions.” *IEEE Transactions on Human-Machine Systems*, 2018. doi: 10.1109/THMS.2018.2874188

Peer Reviewed Conference Papers

- C1 **K. Akash**, T. Reid, and N. Jain, “Improving Human-Machine Collaboration Through Transparency-based Feedback — Part II: Control Design and Synthesis.” Proceedings of the 2nd IFAC Conference on Cyber-Physical & Human-Systems, Miami, FL, December 14–15, 2018.
- C2 **K. Akash**, K. Polson, T. Reid, and N. Jain, “Improving Human-Machine Collaboration Through Transparency-based Feedback — Part I: Human Trust and Workload Model.” Proceedings of the 2nd IFAC Conference on Cyber-Physical & Human-Systems, Miami, FL, December 14–15, 2018.
- C3 **K. Akash**, T. Reid, and N. Jain, “Adaptive Probabilistic Classification of Dynamic Processes: A Case Study on Human Trust in Automation.” Proceedings of the 2018 American Control Conference, Milwaukee, WI, June 27–29, 2018.
- C4 **K. Akash**, W.-L. Hu, T. Reid, and N. Jain, “Dynamic Modeling of Trust in Human-Machine Interactions.” Proceedings of the 2017 American Control Conference, Seattle, WA, May 24–26, 2017.
- C5 W.-L. Hu, **K. Akash**, N. Jain, and T. Reid, “Real-Time Sensing of Trust in Human-Machine Interactions.” Proceedings of the 1st IFAC Conference on Cyber-Physical & Human-Systems, Florianopolis, Brazil, December 7–9, 2016.

Publications (Other)

- O1 **K. Akash**, K. P. Lijesh, V. Chittlangia, and H. Hirani, “Design and Implementation of Adaptive PID Controller for Active Magnetic Bearings.” *Technology Letters*, vol. 1(12), pp. 18-24, 2014.
- O2 V. Chittlangia, K. P. Lijesh, **K. Akash**, and H. Hirani, “Optimum Design of an Active Magnetic Bearing Considering the Geometric Programming.” *Technology Letters*, vol. 1(3), pp. 23-30, 2014.

Patents

- P1 H. Hirani, **K. Akash**, and H. Gupta, “An Internal Hub Gear Assembly.” Indian Patent 3793/DEL/2015, November 20, 2015. (Published, Awaiting Examination)

Invited Talks

- T1 **K. Akash**, T. Reid, and N. Jain, “A Classification Model for Sensing Human Trust in Machines Using EEG and GSR.” ACM Intelligent User Interfaces (IUI) Conference 2019, Los Angeles, CA, March 16-20, 2019.
- T2 **K. Akash**, T. Reid, and N. Jain, “Reimagining Human-Machine Interactions Through Trust-Based Feedback.” Student Lightning Talks, 2019 Southwest Robotics Symposium, Arizona State University, Tempe, AZ, January 24-25, 2019.

Industrial Experience

1. **Research Intern**, Honda Research Institute, San Jose, CA, USA May 2019–August 2019
Adaptive Transparency Framework for Level 2 Autonomous Driving
 - Analyzed the effects of augmented reality-based transparency cues on driver’s cognitive states using eye-tracking, galvanic skin response, and manual takeover tendencies.
 - Developed models to capture the dynamic effects of transparency on driver’s cognitive states.
 - Established optimal control policies to improve driving performance by dynamically varying transparency based on driver’s estimated cognitive states.
2. **Mechanical Intern**, Dover India Innovation Center, Bangalore, India May 2014–July 2014
Designing a 15000 lbf Hydraulic Planetary Winch for Tulsa Winch Group, Oklahoma, USA
 - Reinforced design decisions using structural analysis of critical components to optimize the weight-to-strength trade-off.
 - Designed hydraulic circuit of the winch along with selection and validation of hydraulic motor required to power the hydraulic winch.

Research Experience

1. **Graduate Research Assistant**, Purdue University, IN, USA Aug 2015–Present
Reimagining Human-Machine Interactions through Trust-based Feedback
 - Designed multiple human subject studies to analyze human trust and workload behavior during interactions with an automated decision-aid and collected data using in-person experiments as well as using online experiments conducted through Amazon Mechanical Turk (paper J1, J2, J3, C2, C4).
 - Created machine-learning-based and control-oriented models to estimate and predict human trust and workload based on human behavior (paper J1, J3, C2, C4).
 - Developed a classification-based framework to estimate human trust using extracted features from psychophysiological signals including electroencephalogram (EEG) and galvanic skin response (GSR) (paper J2, C3, C5).
 - Synthesized optimal control algorithms that enable machines to respond to changes in human trust in real time to improve human-machine collaboration and validated closed-loop performance through human subject experimentation (paper J1, C1).
2. **Undergraduate Researcher**, IIT Delhi, India Jul 2014–May 2015
Growth Plate Preserving Intramedullary Nail for Pediatric Patients
 - Modeled and simulated fractured pediatric femur bone.
 - Designed a segmented nailing solution with sufficient rigidity to support human body weight.
 - Analyzed the stability of the bone and implant system, followed by prototyping and testing for design validation.

3. **Undergraduate Researcher**, IIT Delhi, India Jan 2014–Jan 2015
Design of Internal Hub Gear for Bicycles
 - Developed a robust and economical 2-speed hub gear system for Hero Cycles Limited, India's largest cycle manufacturer, to be used in cycles for rural areas.
 - Conceptualized two separate designs based on epicyclic gears; one to be mounted on the rear-wheel and the other on pedals (patent P1).
 - Validated the designs by simulations followed by prototyping and testing.
4. **Undergraduate Researcher**, IIT Delhi, India Jan 2013–May 2014
Design and Development of Active Magnetic Bearing System
 - Developed a frictionless and lubricant free bearing and its controller by implementing a magnetically levitated rotor.
 - Optimized the core design of the electromagnet and developed a prototype of the active magnetic bearing system (paper O2).
 - Designed a Result Adaptive PID controller algorithm that could be heuristically tuned and validated it using a National Instruments Real-Time and FPGA controller. (paper O1)

Mentorship Experience

1. **Neural Network Models to Classify Trust using EEG**, Purdue University Jan 2020–Present
Mentored an undergraduate researcher to develop neural network models for real time estimation of human trust in automation using electroencephalogram (EEG) measurements.
2. **Clustering Partially-Observable Markov Models**, Purdue University Aug 2019–Present
Mentored an undergraduate researcher to analyze and cluster human trust behavior data using unsupervised machine learning techniques.
3. **Medium Fidelity Driving Simulator Setup**, Purdue University Aug 2019–Dec 2019
Worked with two undergraduate researchers to build and program a driving simulator capable of automated driving and human data collection including eye-tracking and electroencephalogram (EEG).
4. **Adaptive Transparency based on Trust & Workload**, Purdue University Aug 2017–Dec 2018
Worked with an undergraduate researcher to develop closed loop policies that can vary automation transparency in real time based on human trust and workload. The control policies were validated and the results are in review to be published in a journal article.
5. **User Interface Design for Analyzing Trust in HMI**, Purdue University May 2016–May 2018
Mentored an undergraduate researcher to design user-interfaces for human subject studies and collect data to analyze and model human trust behavior during human-machine interactions (HMI). The work was presented in a peer-reviewed conference publication.

Professional Activities

Journal Reviewer

- IEEE Transactions on Control Systems Technology 2019–Present
- IEEE Transactions on Human-Machine Systems 2017–Present
- IEEE Access 2017–Present

Conference Reviewer

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| • IFAC Conference on Cyber-Physical & Human Systems (CPHS) | 2018–Present |
| • American Control Conference (ACC) | 2016–Present |
| • IEEE Conference on Decision and Control (CDC) | 2016–Present |

Professional Society Memberships

- American Society of Mechanical Engineers (ASME)
- Institute of Electrical and Electronics Engineers (IEEE)